Resolve.

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Database design

Schema, relationships and data dictionary

Complete SQLite fields, relationships, constraints and migrations, with PostgreSQL migration instructions. Schema inspection uses new disposable storage.

COMPILATION DATE

12 November 2026

Source chapters retain their recorded dates and validation scope. Local, simulated and account-backed evidence are distinguished. This volume does not report a new production acceptance run.

Storage design

Resolve persists tenant-owned operational state separately from the fact that a provider action completed. SQLite is the local default. PostgreSQL uses the shared repository with a dedicated schema and bounded connection pool.

The dictionary was derived from a brand-new disposable SQLiteStore using the current immutable migrations. No existing owner database or hosted storage was opened. It establishes the fresh SQLite schema shape, not PostgreSQL runtime equivalence or production acceptance.

Transactions and financial identity

SQLite enables foreign keys and uses BEGIN IMMEDIATE for write transactions. PostgreSQL retains repeatable-read/read-only reads and transactional write locking. Financial amount_cents fields use integer minor units. Capture histories, cases and refundable capacity are scoped to the merchant.

Approval and execution are distinct records. Saved provider request identities, optimistic revisions, deduplication constraints and worker leases keep retries within the authorized operation. Application validation enforces several business rules beyond physical foreign keys.

Measurement	Observed result
SQLite schema version	28
Tables	70
Columns	537
Foreign-key rows	137
PRAGMA foreign_key_check	passed

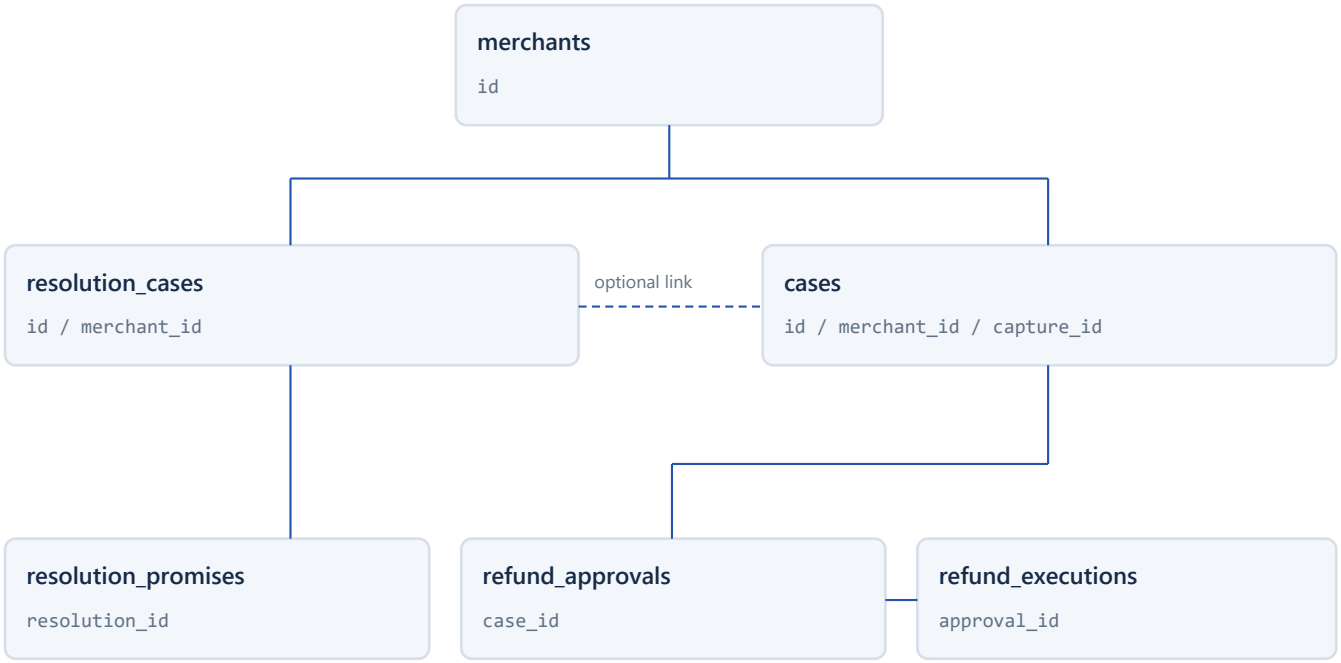
Relationship guide

merchants is the tenant root. Payment cases and resolution cases retain ownership. Promises, evidence, buyer access, commerce snapshots, approvals and execution histories attach to those contexts. Platform, aftercare and assurance records extend staff operations, returns and dispute holds. The table-specific foreign-key lists below are authoritative for physical relationships; not all logical links are foreign keys.

Migration policy

Numbered migrations are immutable. Corrections belong in a new migration. The SQLite initializer applies investigation_prerequisite.sql at its transitional point. PostgreSQL uses a separate migration sequence; those files are inspected here, not executed.

Core relational map



Selected foreign-key links; composite ownership keys are simplified here.

Solid lines show selected foreign-key relationships. The dashed line represents the resolution case's optional refund-case link. This overview omits supporting tables and simplifies composite keys; the following catalog and dictionary retain the full physical schema.

Migration catalog

Engine	Migration	SHA-256 prefix
SQLite	001_initial.sql	7575311879e0
SQLite	002_merchants.sql	832a14549162
SQLite	003_refund_approvals.sql	1da5c4a3a268
SQLite	004_capture_verifications.sql	58c9fb1397ba
SQLite	005_refund_executions.sql	c9aa6f6449ca
SQLite	006_execution_identity.sql	cb1282fe8f69
SQLite	007_case_events.sql	960d27d610a4
SQLite	008_refund_history_and_case_lifecycle.sql	c60f2891d76d
SQLite	009_resolution_workspace.sql	0f652ade8638
SQLite	010_customer_portal.sql	36bc941ba305
SQLite	011_customer_intake_and_triage.sql	32ac598c3b31
SQLite	012_hosted_followup.sql	dffc808f35ae
SQLite	013_case_order.sql	768a0a08c9af
SQLite	014_merchant_web_auth.sql	eb6e217bdade
SQLite	015_merchant_paypal_flow_claim.sql	44ea1956cbd2
SQLite	016_customer_google_and_smtp.sql	f35198f84e02
SQLite	017_customer_email_notifications.sql	c6c2a60a4928
SQLite	018_resolution_evidence_sdk.sql	60d75405896d
SQLite	019_commerce_evidence.sql	39b5d2b196be
SQLite	020_merchant_orders.sql	1cb822a224f2
SQLite	021_investigations.sql	13effb5f4329
SQLite	022_investigation_commerce.sql	b88a66f61ffe
SQLite	023_investigation_payment_facts.sql	502a099e928c
SQLite	024_investigation_financial_state.sql	702f2a2702c4
SQLite	025_investigation_queue_audit.sql	25c8e7ee46b5
SQLite	026_merchant_operations.sql	0a7aa42cbada
SQLite	027_embedded_aftercare.sql	f10b62d01669
SQLite	028_resolution_assurance.sql	67f49aeadd3f
SQLite	investigation_prerequisite.sql	a0039a210380
PostgreSQL	001_baseline.sql	4d1ed4281568
PostgreSQL	002_case_order.sql	35d9a244575d

Engine	Migration	SHA-256 prefix
PostgreSQL	003_merchant_web_auth.sql	eb6e217bdade
PostgreSQL	004_merchant_paypal_flow_claim.sql	4c958258bd16
PostgreSQL	005_customer_google_and_smtp.sql	f35198f84e02
PostgreSQL	006_customer_email_notifications.sql	6bd216c8fce4
PostgreSQL	007_resolution_evidence_sdk.sql	60d75405896d
PostgreSQL	008_commerce_evidence.sql	39b5d2b196be
PostgreSQL	009_merchant_orders.sql	1cb822a224f2
PostgreSQL	010_investigations.sql	2385248e8089
PostgreSQL	011_investigation_commerce.sql	9a1d0f7ef1b7
PostgreSQL	012_investigation_payment_facts.sql	52eb16dc5de2
PostgreSQL	013_investigation_financial_state.sql	bc55785aa84c
PostgreSQL	014_investigation_import_constraints.sql	337e77ce61ca
PostgreSQL	015_investigation_queue_audit.sql	8da71000bcd4
PostgreSQL	016_merchant_operations.sql	0a7aa42cbada
PostgreSQL	017_operations_import_constraints.sql	04698b4146be
PostgreSQL	018_embedded_aftercare.sql	53bf2e50cb77
PostgreSQL	019_resolution_assurance.sql	a71abaeefb6f

Table catalog

Table	Columns	FK rows
aftercare_apps	4	2
aftercare_issues	7	3
assurance_dispute_events	9	1
assurance_disputes	9	1
assurance_events	9	2
assurance_policies	4	1
assurance_projections	6	2
assurance_receipts	4	2
assurance_refund_traces	8	1
assurance_return_guides	5	2
assurance_return_labels	7	2
assurance_return_reviews	8	2
captures	4	1
case_attachments	10	3
case_events	8	3
cases	11	3
commerce_orders	9	3
condition_checks	6	2
customer_account_links	5	2
customer_case_contacts	5	2
customer_evidence	9	2
customer_google_flows	4	0
customer_intake_invites	10	3
customer_mail_outbox	13	1
customer_portal_links	8	2
customer_portal_messages	8	2
customer_sessions	3	0
customer_signins	5	0
investigation_audit	6	2
investigation_jobs	8	2
investigation_reviews	3	2

Table	Columns	FK rows
investigations	8	2
merchant_api_keys	6	1
merchant_paypal_flows	7	1
merchant_paypal_links	5	1
merchant_web_sessions	6	3
merchants	3	0
paypal_capture_verifications	10	3
paypal_refund_history	8	2
paypal_refund_verifications	11	4
paypal_webhook_events	10	1
platform_alerts	12	2
platform_audit	7	1
platform_auth_limits	3	0
platform_connections	9	1
platform_imports	11	2
platform_invites	8	1
platform_offers	11	2
platform_packages	10	2
platform_returns	13	3
platform_sessions	4	1
platform_staff	8	1
platform_ticket_messages	7	1
platform_webhook_jobs	12	1
provider_refresh_schedule	5	1
refund_approval_events	6	2
refund_approvals	17	4
refund_execution_events	6	1
refund_executions	21	7
refunds	7	3
resolution_cases	6	3
resolution_events	10	5
resolution_evidence_events	7	3
resolution_evidence_links	7	2

Table	Columns	FK rows
resolution_evidence_refs	4	2
resolution_evidence_state	4	2
resolution_helpdesk	6	2
resolution_operations	7	2
resolution_promises	9	2
review_exceptions	11	4

Data dictionary

Nullability is reported by PRAGMA table_info. SQLite may report notnull=0 for a primary key; primary-key position is listed separately. Stored definitions preserve CHECK/UNIQUE clauses not fully described by the column list. Defaults are schema expressions, not customer values.

aftercare_apps

Column	Storage type	Null / key / default
key_id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
source	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION
key_id	merchant_api_keys.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_aftercare_apps_1 | UNIQUE | key_id

Stored table definition

```
CREATE TABLE aftercare_apps (
 key_id TEXT PRIMARY KEY REFERENCES merchant_api_keys(id),
 merchant_id TEXT NOT NULL REFERENCES merchants(id),
 source TEXT NOT NULL, created_at TEXT NOT NULL
)
```

aftercare_issues

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
order_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
request_id	TEXT	NOT NULL
request_json	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
case_id	resolution_cases.id	NO ACTION / NO ACTION
order_id	commerce_orders.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

aftercare_issues_case_idx | index | merchant_id, case_id

sqlite_autoindex_aftercare_issues_2 | UNIQUE | merchant_id, request_id

sqlite_autoindex_aftercare_issues_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE aftercare_issues (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 order_id TEXT NOT NULL REFERENCES commerce_orders(id),
 case_id TEXT NOT NULL REFERENCES resolution_cases(id),
 request_id TEXT NOT NULL, request_json TEXT NOT NULL, created_at TEXT NOT NULL,
 UNIQUE(merchant_id,request_id)
)
```

assurance_dispute_events

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
provider	TEXT	NOT NULL
event_id	TEXT	NOT NULL
body_hash	TEXT	NOT NULL
reference	TEXT	NOT NULL
version	INTEGER	NOT NULL
outcome	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_dispute_events_2 | UNIQUE | merchant_id, provider, event_id

sqlite_autoindex_assurance_dispute_events_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE assurance_dispute_events (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id), provider TEXT NOT NULL,
 event_id TEXT NOT NULL, body_hash TEXT NOT NULL, reference TEXT NOT NULL, version INTEGER NOT NULL,
 outcome TEXT NOT NULL, created_at TEXT NOT NULL, UNIQUE(merchant_id,provider,event_id)
)
```

assurance_disputes

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
provider	TEXT	NOT NULL
reference	TEXT	NOT NULL
version	INTEGER	NOT NULL
state	TEXT	NOT NULL
data_json	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_disputes_2 | UNIQUE | merchant_id, provider, reference

sqlite_autoindex_assurance_disputes_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE assurance_disputes (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id), capture_id TEXT NOT NULL,
 provider TEXT NOT NULL, reference TEXT NOT NULL, version INTEGER NOT NULL,
 state TEXT NOT NULL, data_json TEXT NOT NULL, updated_at TEXT NOT NULL,
 UNIQUE(merchant_id,provider,reference)
)
```

assurance_events

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
source	TEXT	NOT NULL
order_id	TEXT	NOT NULL
order_reference	TEXT	NOT NULL
revision	INTEGER	NOT NULL
snapshot_json	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
order_id	commerce_orders.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

assurance_events_scope_idx | index | merchant_id, source, sequence

sqlite_autoindex_assurance_events_2 | UNIQUE | order_id, revision

sqlite_autoindex_assurance_events_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE assurance_events (
 sequence INTEGER PRIMARY KEY AUTOINCREMENT, id TEXT NOT NULL UNIQUE,
 merchant_id TEXT NOT NULL REFERENCES merchants(id), source TEXT NOT NULL,
 order_id TEXT NOT NULL REFERENCES commerce_orders(id), order_reference TEXT NOT NULL,
 revision INTEGER NOT NULL, snapshot_json TEXT NOT NULL, created_at TEXT NOT NULL,
 UNIQUE(order_id,revision)
)
```

assurance_policies

Column	Storage type	Null / key / default
merchant_id	TEXT	nullable per pragma; PK position 1
revision	INTEGER	NOT NULL
data_json	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_policies_1 | UNIQUE | merchant_id

Stored table definition

```
CREATE TABLE assurance_policies (
 merchant_id TEXT PRIMARY KEY REFERENCES merchants(id), revision INTEGER NOT NULL,
 data_json TEXT NOT NULL, updated_at TEXT NOT NULL
)
```

assurance_projections

Column	Storage type	Null / key / default
order_id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
revision	INTEGER	NOT NULL
fingerprint	TEXT	NOT NULL
snapshot_json	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION
order_id	commerce_orders.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_projections_1 | UNIQUE | order_id

Stored table definition

```
CREATE TABLE assurance_projections (
 order_id TEXT PRIMARY KEY REFERENCES commerce_orders(id), merchant_id TEXT NOT NULL REFERENCES merchants(id),
 revision INTEGER NOT NULL, fingerprint TEXT NOT NULL, snapshot_json TEXT NOT NULL, updated_at TEXT NOT NULL
)
```

assurance_receipts

Column	Storage type	Null / key / default
resolution_id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
received_at	TEXT	NOT NULL
source	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION
resolution_id	resolution_cases.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_receipts_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE assurance_receipts (
 resolution_id TEXT PRIMARY KEY REFERENCES resolution_cases(id), merchant_id TEXT NOT NULL REFERENCES merchants(id),
 received_at TEXT NOT NULL, source TEXT NOT NULL
)
```

assurance_refund_traces

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
refund_id	TEXT	NOT NULL
request_id	TEXT	NOT NULL
data_json	TEXT	NOT NULL
actor_id	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_refund_traces_2 | UNIQUE | merchant_id, request_id

sqlite_autoindex_assurance_refund_traces_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE assurance_refund_traces (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id), capture_id TEXT NOT NULL,
 refund_id TEXT NOT NULL, request_id TEXT NOT NULL, data_json TEXT NOT NULL,
 actor_id TEXT NOT NULL, created_at TEXT NOT NULL, UNIQUE(merchant_id,request_id)
)
```

assurance_return_guides

Column	Storage type	Null / key / default
return_id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
revision	INTEGER	NOT NULL
data_json	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION
return_id	platform_returns.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_return_guides_1 | UNIQUE | return_id

Stored table definition

```
CREATE TABLE assurance_return_guides (
 return_id TEXT PRIMARY KEY REFERENCES platform_returns(id), merchant_id TEXT NOT NULL REFERENCES merchants(id),
 revision INTEGER NOT NULL, data_json TEXT NOT NULL, updated_at TEXT NOT NULL
)
```

assurance_return_labels

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
return_id	TEXT	NOT NULL
filename	TEXT	NOT NULL
content	BLOB	NOT NULL
sha256	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
return_id	platform_returns.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_return_labels_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE assurance_return_labels (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id), return_id TEXT NOT NULL REFERENCES
platform_returns(id),
 filename TEXT NOT NULL, content BLOB NOT NULL, sha256 TEXT NOT NULL, created_at TEXT NOT NULL
)
```

assurance_return_reviews

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
return_id	TEXT	NOT NULL
request_id	TEXT	NOT NULL
kind	TEXT	NOT NULL
data_json	TEXT	NOT NULL
actor_id	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
return_id	platform_returns.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_assurance_return_reviews_2 | UNIQUE | merchant_id, request_id

sqlite_autoindex_assurance_return_reviews_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE assurance_return_reviews (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id), return_id TEXT NOT NULL REFERENCES
platform_returns(id),
 request_id TEXT NOT NULL, kind TEXT NOT NULL, data_json TEXT NOT NULL, actor_id TEXT NOT NULL, created_at TEXT NOT NULL,
 UNIQUE(merchant_id,request_id)
)
```

captures

Column	Storage type	Null / key / default
merchant_id	TEXT	NOT NULL; PK position 1
capture_id	TEXT	NOT NULL; PK position 2
captured_cents	INTEGER	NOT NULL
currency	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_captures_1 | UNIQUE | merchant_id, capture_id

Stored table definition

```
CREATE TABLE captures (
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    capture_id TEXT NOT NULL,
    captured_cents INTEGER NOT NULL CHECK (captured_cents > 0),
    currency TEXT NOT NULL CHECK (currency = 'USD'),
    PRIMARY KEY (merchant_id, capture_id)
)
```

case_attachments

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
filename	TEXT	NOT NULL
content_type	TEXT	NOT NULL
size_bytes	INTEGER	NOT NULL
sha256	TEXT	NOT NULL
content	BLOB	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
capture_id	cases.capture_id	CASCADE / NO ACTION

Indexes and uniqueness

case_attachments_owner_idx | index | merchant_id, case_id, created_at

sqlite_autoindex_case_attachments_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE case_attachments (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    case_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    filename TEXT NOT NULL,
    content_type TEXT NOT NULL,
    size_bytes INTEGER NOT NULL CHECK (size_bytes BETWEEN 1 AND 2097152),
    sha256 TEXT NOT NULL,
    content BLOB NOT NULL CHECK (length(content) = size_bytes),
    created_at TEXT NOT NULL,
    FOREIGN KEY (case_id, merchant_id, capture_id)
        REFERENCES cases(id, merchant_id, capture_id) ON UPDATE CASCADE
)
```

case_events

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
event_type	TEXT	NOT NULL
payload_json	TEXT	NOT NULL
recorded_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
capture_id	cases.capture_id	CASCADE / NO ACTION

Indexes and uniqueness

case_events_capture_idx | index | merchant_id, capture_id, sequence

sqlite_autoindex_case_events_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE case_events (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    id TEXT NOT NULL UNIQUE,
    merchant_id TEXT NOT NULL,
    case_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    event_type TEXT NOT NULL,
    payload_json TEXT NOT NULL,
    recorded_at TEXT NOT NULL,
    FOREIGN KEY (case_id, merchant_id, capture_id)
        REFERENCES cases(id, merchant_id, capture_id) ON UPDATE CASCADE
)
```

cases

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
order_id	TEXT	NOT NULL
complaint	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
commitment_json	TEXT	nullable per pragma
status	TEXT	NOT NULL
condition_epoch	TEXT	nullable per pragma
closed_at	TEXT	nullable per pragma
lifecycle_revision	INTEGER	NOT NULL; default 0
case_sequence	INTEGER	nullable per pragma

Foreign keys

Local column	Target	Update / delete
merchant_id	captures.merchant_id	CASCADE / NO ACTION
capture_id	captures.capture_id	CASCADE / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

cases_sequence_idx | UNIQUE | case_sequence

cases_resolution_owner_idx | UNIQUE | id, merchant_id

cases_approval_owner_idx | UNIQUE | id, merchant_id, capture_id

cases_merchant_capture_idx | index | merchant_id, capture_id

sqlite_autoindex_cases_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE cases (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    order_id TEXT NOT NULL,
    complaint TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    commitment_json TEXT,
    status TEXT NOT NULL CHECK (status = 'open'), condition_epoch TEXT, closed_at TEXT, lifecycle_revision INTEGER NOT
NULL DEFAULT 0, case_sequence INTEGER,
    FOREIGN KEY (merchant_id, capture_id)
        REFERENCES captures(merchant_id, capture_id) ON UPDATE CASCADE
)
```

commerce_orders

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
resolution_id	TEXT	nullable per pragma
merchant_id	TEXT	NOT NULL
source	TEXT	NOT NULL
order_reference	TEXT	NOT NULL
revision	BIGINT	NOT NULL
snapshot_json	TEXT	NOT NULL
contact_json	TEXT	nullable per pragma
synced_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

commerce_orders_merchant_recent | index | merchant_id, synced_at, id

sqlite_autoindex_commerce_orders_3 | UNIQUE | merchant_id, source, order_reference

sqlite_autoindex_commerce_orders_2 | UNIQUE | resolution_id

sqlite_autoindex_commerce_orders_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE commerce_orders (
    id TEXT PRIMARY KEY NOT NULL,
    resolution_id TEXT UNIQUE,
    merchant_id TEXT NOT NULL REFERENCES merchants(id) DEFERRABLE INITIALLY IMMEDIATE,
    source TEXT NOT NULL,
    order_reference TEXT NOT NULL,
    revision BIGINT NOT NULL CHECK (revision > 0),
    snapshot_json TEXT NOT NULL,
    contact_json TEXT,
    synced_at TEXT NOT NULL,
    UNIQUE (merchant_id, source, order_reference),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) DEFERRABLE INITIALLY IMMEDIATE
)
```

condition_checks

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
case_id	TEXT	NOT NULL
status	TEXT	NOT NULL
evidence_note	TEXT	NOT NULL
recorded_at	TEXT	NOT NULL
recorded_by_merchant_id	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
recorded_by_merchant_id	merchants.id	NO ACTION / NO ACTION
case_id	cases.id	NO ACTION / NO ACTION

Indexes and uniqueness

condition_checks_case_idx | index | case_id

Stored table definition

```
CREATE TABLE condition_checks (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    case_id TEXT NOT NULL REFERENCES cases(id),
    status TEXT NOT NULL CHECK (status IN ('met', 'not_met')),
    evidence_note TEXT NOT NULL,
    recorded_at TEXT NOT NULL,
    recorded_by_merchant_id TEXT REFERENCES merchants(id)
)
```

customer_account_links

Column	Storage type	Null / key / default
token_hash	TEXT	NOT NULL; PK position 1
session_hash	TEXT	NOT NULL
resolution_id	TEXT	NOT NULL
parent_hash	TEXT	NOT NULL
expires_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	customer_case_contacts.resolution_id	NO ACTION / NO ACTION
session_hash	customer_sessions.token_hash	NO ACTION / CASCADE

Indexes and uniqueness

sqlite_autoindex_customer_account_links_1 | UNIQUE | token_hash

Stored table definition

```
CREATE TABLE customer_account_links (
    token_hash TEXT PRIMARY KEY NOT NULL,
    session_hash TEXT NOT NULL REFERENCES customer_sessions(token_hash) ON DELETE CASCADE,
    resolution_id TEXT NOT NULL REFERENCES customer_case_contacts(resolution_id),
    parent_hash TEXT NOT NULL,
    expires_at TEXT NOT NULL
)
```

customer_case_contacts

Column	Storage type	Null / key / default
resolution_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
email	TEXT	NOT NULL
created_at	TEXT	NOT NULL
email_notifications	INTEGER	NOT NULL; default 1

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

customer_contacts_email_idx | index | email, resolution_id

sqlite_autoindex_customer_case_contacts_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE customer_case_contacts (
    resolution_id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    email TEXT NOT NULL,
    created_at TEXT NOT NULL, email_notifications INTEGER NOT NULL DEFAULT 1 CHECK (email_notifications IN (0,1)),
    FOREIGN KEY (resolution_id,merchant_id) REFERENCES resolution_cases(id,merchant_id)
)
```

customer_evidence

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
resolution_id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
filename	TEXT	NOT NULL
content_type	TEXT	NOT NULL
content	BLOB	NOT NULL
size_bytes	INTEGER	NOT NULL
sha256	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	CASCADE / NO ACTION
merchant_id	resolution_cases.merchant_id	CASCADE / NO ACTION

Indexes and uniqueness

customer_evidence_case_idx | index | resolution_id, created_at, id

sqlite_autoindex_customer_evidence_2 | UNIQUE | resolution_id, filename, sha256

sqlite_autoindex_customer_evidence_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE customer_evidence (
    id TEXT PRIMARY KEY NOT NULL,
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    filename TEXT NOT NULL,
    content_type TEXT NOT NULL,
    content BLOB NOT NULL,
    size_bytes INTEGER NOT NULL CHECK(size_bytes BETWEEN 1 AND 2097152),
    sha256 TEXT NOT NULL,
    created_at TEXT NOT NULL,
    UNIQUE (resolution_id, filename, sha256),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) ON UPDATE CASCADE
)
```

customer_google_flows

Column	Storage type	Null / key / default
state_hash	TEXT	NOT NULL; PK position 1
browser_hash	TEXT	NOT NULL
app_key	TEXT	NOT NULL
expires_at	TEXT	NOT NULL

Indexes and uniqueness

customer_google_flow_expiry_idx | index | expires_at

sqlite_autoindex_customer_google_flows_1 | UNIQUE | state_hash

Stored table definition

```
CREATE TABLE customer_google_flows (
    state_hash TEXT PRIMARY KEY NOT NULL,
    browser_hash TEXT NOT NULL,
    app_key TEXT NOT NULL,
    expires_at TEXT NOT NULL
)
```

customer_intake_invites

Column	Storage type	Null / key / default
token_hash	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
order_reference	TEXT	NOT NULL
created_at	TEXT	NOT NULL
expires_at	TEXT	NOT NULL
resolution_id	TEXT	nullable per pragma
submitted_at	TEXT	nullable per pragma
issue_type	TEXT	nullable per pragma
customer_email	TEXT	nullable per pragma
email_notifications	INTEGER	NOT NULL; default 1

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	CASCADE / NO ACTION
merchant_id	resolution_cases.merchant_id	CASCADE / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_customer_intake_invites_2 | UNIQUE | resolution_id

sqlite_autoindex_customer_intake_invites_1 | UNIQUE | token_hash

Stored table definition

```
CREATE TABLE customer_intake_invites (
    token_hash TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    order_reference TEXT NOT NULL CHECK(length(order_reference) BETWEEN 1 AND 160),
    created_at TEXT NOT NULL,
    expires_at TEXT NOT NULL,
    resolution_id TEXT UNIQUE,
    submitted_at TEXT,
    issue_type TEXT, customer_email TEXT, email_notifications INTEGER NOT NULL DEFAULT 1 CHECK (email_notifications IN
(0,1)),
    CHECK ((resolution_id IS NULL) = (submitted_at IS NULL)),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) ON UPDATE CASCADE
)
```

customer_mail_outbox

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
event_key	TEXT	NOT NULL
kind	TEXT	NOT NULL
recipient	TEXT	NOT NULL
resolution_id	TEXT	nullable per pragma
created_at	TEXT	NOT NULL
expires_at	TEXT	NOT NULL
next_attempt_at	TEXT	NOT NULL
attempts	INTEGER	NOT NULL; default 0
lease_id	TEXT	nullable per pragma
status	TEXT	NOT NULL; default 'queued'
provider_id	TEXT	nullable per pragma
delivery_transport	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION

Indexes and uniqueness

customer_mail_due_idx | index | status, next_attempt_at

sqlite_autoindex_customer_mail_outbox_2 | UNIQUE | event_key

sqlite_autoindex_customer_mail_outbox_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE customer_mail_outbox (
    id TEXT PRIMARY KEY NOT NULL,
    event_key TEXT NOT NULL UNIQUE,
    kind TEXT NOT NULL CHECK(kind IN ('invite','signin','reply','refund')),
    recipient TEXT NOT NULL,
    resolution_id TEXT REFERENCES resolution_cases(id),
    created_at TEXT NOT NULL,
    expires_at TEXT NOT NULL,
    next_attempt_at TEXT NOT NULL,
    attempts INTEGER NOT NULL DEFAULT 0,
    lease_id TEXT,
    status TEXT NOT NULL DEFAULT 'queued' CHECK(status IN ('queued','accepted','failed','cancelled')),
    provider_id TEXT
, delivery_transport TEXT CHECK (delivery_transport IN ('resend','smtp')))
```

customer_portal_links

Column	Storage type	Null / key / default
resolution_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
token_hash	TEXT	NOT NULL
title	TEXT	NOT NULL
show_order_reference	INTEGER	NOT NULL
created_at	TEXT	NOT NULL
expires_at	TEXT	NOT NULL
revoked_at	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	CASCADE / NO ACTION
merchant_id	resolution_cases.merchant_id	CASCADE / NO ACTION

Indexes and uniqueness

sqlite_autoindex_customer_portal_links_2 | UNIQUE | token_hash

sqlite_autoindex_customer_portal_links_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE customer_portal_links (
    resolution_id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    token_hash TEXT NOT NULL UNIQUE,
    title TEXT NOT NULL CHECK (length(title) BETWEEN 1 AND 120),
    show_order_reference INTEGER NOT NULL CHECK (show_order_reference IN (0, 1)),
    created_at TEXT NOT NULL,
    expires_at TEXT NOT NULL,
    revoked_at TEXT,
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) ON UPDATE CASCADE
)
```

customer_portal_messages

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
id	TEXT	NOT NULL
resolution_id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
sender	TEXT	NOT NULL
request_id	TEXT	NOT NULL
body	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	CASCADE / NO ACTION
merchant_id	resolution_cases.merchant_id	CASCADE / NO ACTION

Indexes and uniqueness

customer_portal_messages_rate_idx | index | resolution_id, sender, created_at

customer_portal_messages_case_idx | index | resolution_id, sequence

sqlite_autoindex_customer_portal_messages_2 | UNIQUE | resolution_id, sender, request_id

sqlite_autoindex_customer_portal_messages_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE customer_portal_messages (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    id TEXT NOT NULL UNIQUE,
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    sender TEXT NOT NULL CHECK (sender IN ('customer', 'merchant')),
    request_id TEXT NOT NULL,
    body TEXT NOT NULL CHECK (length(body) BETWEEN 1 AND 2000),
    created_at TEXT NOT NULL,
    UNIQUE (resolution_id, sender, request_id),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) ON UPDATE CASCADE
)
```

customer_sessions

Column	Storage type	Null / key / default
token_hash	TEXT	NOT NULL; PK position 1
email	TEXT	NOT NULL
expires_at	TEXT	NOT NULL

Indexes and uniqueness

sqlite_autoindex_customer_sessions_1 | UNIQUE | token_hash

Stored table definition

```
CREATE TABLE customer_sessions (
    token_hash TEXT PRIMARY KEY NOT NULL,
    email TEXT NOT NULL,
    expires_at TEXT NOT NULL
)
```

customer_signins

Column	Storage type	Null / key / default
token_hash	TEXT	NOT NULL; PK position 1
email	TEXT	NOT NULL
created_at	TEXT	NOT NULL
expires_at	TEXT	NOT NULL
consumed_at	TEXT	nullable per pragma

Indexes and uniqueness

customer_signins_email_idx | index | email, created_at

sqlite_autoindex_customer_signins_1 | UNIQUE | token_hash

Stored table definition

```
CREATE TABLE customer_signins (
    token_hash TEXT PRIMARY KEY NOT NULL,
    email TEXT NOT NULL,
    created_at TEXT NOT NULL,
    expires_at TEXT NOT NULL,
    consumed_at TEXT
)
```

investigation_audit

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
resolution_id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
recorded_at	TEXT	NOT NULL
event_type	TEXT	NOT NULL
data_json	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_investigation_audit_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE investigation_audit (
    id TEXT PRIMARY KEY NOT NULL,
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    recorded_at TEXT NOT NULL,
    event_type TEXT NOT NULL,
    data_json TEXT NOT NULL,
    FOREIGN KEY(resolution_id,merchant_id) REFERENCES resolution_cases(id,merchant_id)
)
```

investigation_jobs

Column	Storage type	Null / key / default
resolution_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
revision	BIGINT	NOT NULL; default 1
state	TEXT	NOT NULL; default 'queued'
attempts	INTEGER	NOT NULL; default 0
lease_token	TEXT	nullable per pragma
lease_until	TEXT	nullable per pragma
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_investigation_jobs_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE investigation_jobs (
    resolution_id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    revision BIGINT NOT NULL DEFAULT 1,
    state TEXT NOT NULL DEFAULT 'queued' CHECK(state IN ('queued','running','finished','blocked')),
    attempts INTEGER NOT NULL DEFAULT 0,
    lease_token TEXT,
    lease_until TEXT,
    created_at TEXT NOT NULL,
    FOREIGN KEY(resolution_id,merchant_id) REFERENCES resolution_cases(id,merchant_id)
)
```

investigation_reviews

Column	Storage type	Null / key / default
investigation_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
receipt_json	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION
investigation_id	investigations.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_investigation_reviews_1 | UNIQUE | investigation_id

Stored table definition

```
CREATE TABLE investigation_reviews (
    investigation_id TEXT PRIMARY KEY NOT NULL REFERENCES investigations(id),
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    receipt_json TEXT NOT NULL
)
```

investigations

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
resolution_id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
revision	BIGINT	NOT NULL
state	TEXT	NOT NULL
created_at	TEXT	NOT NULL
completed_at	TEXT	NOT NULL
result_json	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

investigations_owner | index | merchant_id, resolution_id, revision

sqlite_autoindex_investigations_2 | UNIQUE | resolution_id, revision

sqlite_autoindex_investigations_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE investigations (
    id TEXT PRIMARY KEY NOT NULL,
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    revision BIGINT NOT NULL,
    state TEXT NOT NULL CHECK(state IN ('finished','blocked')),
    created_at TEXT NOT NULL,
    completed_at TEXT NOT NULL,
    result_json TEXT NOT NULL,
    UNIQUE(resolution_id,revision),
    FOREIGN KEY(resolution_id,merchant_id) REFERENCES resolution_cases(id,merchant_id)
)
```

merchant_api_keys

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
token_hash	TEXT	NOT NULL
created_at	TEXT	NOT NULL
revoked_at	TEXT	nullable per pragma
scope	TEXT	NOT NULL; default 'full'

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_merchant_api_keys_2 | UNIQUE | token_hash

sqlite_autoindex_merchant_api_keys_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE merchant_api_keys (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    token_hash TEXT UNIQUE NOT NULL,
    created_at TEXT NOT NULL,
    revoked_at TEXT
, scope TEXT NOT NULL DEFAULT 'full' CHECK (scope IN ('full','commerce')))
```

merchant_paypal_flows

Column	Storage type	Null / key / default
state_hash	TEXT	NOT NULL; PK position 1
browser_hash	TEXT	NOT NULL
issuer_key	TEXT	NOT NULL
intent	TEXT	NOT NULL
session_hash	TEXT	nullable per pragma
expires_at	TEXT	NOT NULL
claimed_at	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
session_hash	merchant_web_sessions.token_hash	NO ACTION / CASCADE

Indexes and uniqueness

merchant_paypal_flow_expiry_idx | index | expires_at

sqlite_autoindex_merchant_paypal_flows_1 | UNIQUE | state_hash

Stored table definition

```
CREATE TABLE merchant_paypal_flows (
    state_hash TEXT PRIMARY KEY NOT NULL,
    browser_hash TEXT NOT NULL,
    issuer_key TEXT NOT NULL,
    intent TEXT NOT NULL CHECK (intent IN ('signin','link')),
    session_hash TEXT REFERENCES merchant_web_sessions(token_hash) ON DELETE CASCADE,
    expires_at TEXT NOT NULL, claimed_at TEXT,
    CHECK ((intent='link') = (session_hash IS NOT NULL))
)
```

merchant_paypal_links

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
issuer_key	TEXT	NOT NULL
subject_hash	TEXT	NOT NULL
linked_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_merchant_paypal_links_3 | UNIQUE | issuer_key, subject_hash

sqlite_autoindex_merchant_paypal_links_2 | UNIQUE | merchant_id, issuer_key

sqlite_autoindex_merchant_paypal_links_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE merchant_paypal_links (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    issuer_key TEXT NOT NULL,
    subject_hash TEXT NOT NULL,
    linked_at TEXT NOT NULL,
    UNIQUE (merchant_id,issuer_key),
    UNIQUE (issuer_key,subject_hash)
)
```

merchant_web_sessions

Column	Storage type	Null / key / default
token_hash	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
api_key_id	TEXT	nullable per pragma
paypal_link_id	TEXT	nullable per pragma
issued_at	TEXT	NOT NULL
expires_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
paypal_link_id	merchant_paypal_links.id	NO ACTION / CASCADE
api_key_id	merchant_api_keys.id	NO ACTION / CASCADE
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

merchant_web_session_expiry_idx | index | expires_at

sqlite_autoindex_merchant_web_sessions_1 | UNIQUE | token_hash

Stored table definition

```
CREATE TABLE merchant_web_sessions (
    token_hash TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    api_key_id TEXT REFERENCES merchant_api_keys(id) ON DELETE CASCADE,
    paypal_link_id TEXT REFERENCES merchant_paypal_links(id) ON DELETE CASCADE,
    issued_at TEXT NOT NULL,
    expires_at TEXT NOT NULL,
    CHECK ((api_key_id IS NULL) <> (paypal_link_id IS NULL))
)
```

merchants

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
name	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Indexes and uniqueness

sqlite_autoindex_merchants_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE merchants (
    id TEXT PRIMARY KEY NOT NULL,
    name TEXT NOT NULL,
    created_at TEXT NOT NULL
)
```

paypal_capture_verifications

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
paypal_merchant_id	TEXT	NOT NULL
captured_cents	INTEGER	NOT NULL
currency	TEXT	NOT NULL
capture_status	TEXT	NOT NULL
checked_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
capture_id	cases.capture_id	CASCADE / NO ACTION

Indexes and uniqueness

paypal_capture_verifications_capture_idx | index | merchant_id, capture_id, sequence

sqlite_autoindex_paypal_capture_verifications_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE paypal_capture_verifications (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    id TEXT NOT NULL UNIQUE,
    merchant_id TEXT NOT NULL,
    case_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    paypal_merchant_id TEXT NOT NULL,
    captured_cents INTEGER NOT NULL CHECK (captured_cents > 0),
    currency TEXT NOT NULL CHECK (currency = 'USD'),
    capture_status TEXT NOT NULL,
    checked_at TEXT NOT NULL,
    FOREIGN KEY (case_id, merchant_id, capture_id)
        REFERENCES cases(id, merchant_id, capture_id) ON UPDATE CASCADE
)
```

paypal_refund_history

Column	Storage type	Null / key / default
merchant_id	TEXT	NOT NULL; PK position 1
capture_id	TEXT	NOT NULL; PK position 2
revision	TEXT	NOT NULL
digest	TEXT	NOT NULL
order_id	TEXT	NOT NULL
identity_hash	TEXT	NOT NULL
checked_at	TEXT	NOT NULL
snapshot_json	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	captures.merchant_id	CASCADE / NO ACTION
capture_id	captures.capture_id	CASCADE / NO ACTION

Indexes and uniqueness

sqlite_autoindex_paypal_refund_history_1 | UNIQUE | merchant_id, capture_id

Stored table definition

```
CREATE TABLE paypal_refund_history (
    merchant_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    revision TEXT NOT NULL,
    digest TEXT NOT NULL,
    order_id TEXT NOT NULL,
    identity_hash TEXT NOT NULL,
    checked_at TEXT NOT NULL,
    snapshot_json TEXT NOT NULL,
    PRIMARY KEY (merchant_id, capture_id),
    FOREIGN KEY (merchant_id, capture_id) REFERENCES captures(merchant_id, capture_id)
        ON UPDATE CASCADE
)
```

paypal_refund_verifications

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
refund_id	TEXT	NOT NULL
amount_cents	INTEGER	NOT NULL
currency	TEXT	NOT NULL
provider_status	TEXT	NOT NULL
checked_at	TEXT	NOT NULL
execution_id	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
capture_id	cases.capture_id	CASCADE / NO ACTION
execution_id	refund_executions.id	NO ACTION / NO ACTION

Indexes and uniqueness

paypal_refund_verifications_capture_idx | index | merchant_id, capture_id, sequence

sqlite_autoindex_paypal_refund_verifications_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE paypal_refund_verifications (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    id TEXT NOT NULL UNIQUE,
    merchant_id TEXT NOT NULL,
    case_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    refund_id TEXT NOT NULL,
    amount_cents INTEGER NOT NULL CHECK (amount_cents > 0),
    currency TEXT NOT NULL CHECK (currency = 'USD'),
    provider_status TEXT NOT NULL CHECK (provider_status IN ('CANCELLED','FAILED','PENDING','COMPLETED')),
    checked_at TEXT NOT NULL,
    execution_id TEXT REFERENCES refund_executions(id),
    FOREIGN KEY (case_id, merchant_id, capture_id)
        REFERENCES cases(id, merchant_id, capture_id) ON UPDATE CASCADE
)
```

paypal_webhook_events

Column	Storage type	Null / key / default
merchant_id	TEXT	NOT NULL; PK position 1
event_id	TEXT	NOT NULL; PK position 2
payload_sha256	TEXT	NOT NULL
event_type	TEXT	NOT NULL
capture_id	TEXT	nullable per pragma
refund_id	TEXT	nullable per pragma
received_at	TEXT	NOT NULL
status	TEXT	NOT NULL
attempts	INTEGER	NOT NULL; default 0
next_attempt_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

paypal_webhook_capture_idx | index | merchant_id, capture_id, status

paypal_webhook_due_idx | index | merchant_id, status, next_attempt_at

sqlite_autoindex_paypal_webhook_events_1 | UNIQUE | merchant_id, event_id

Stored table definition

```
CREATE TABLE paypal_webhook_events (
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    event_id TEXT NOT NULL,
    payload_sha256 TEXT NOT NULL,
    event_type TEXT NOT NULL,
    capture_id TEXT,
    refund_id TEXT,
    received_at TEXT NOT NULL,
    status TEXT NOT NULL CHECK (status IN ('queued','processed','ignored','exception')),
    attempts INTEGER NOT NULL DEFAULT 0,
    next_attempt_at TEXT NOT NULL,
    PRIMARY KEY (merchant_id, event_id)
)
```

platform_alerts

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
promise_id	TEXT	NOT NULL
promise_revision	INTEGER	NOT NULL
day	TEXT	NOT NULL
state	TEXT	NOT NULL
assignee	TEXT	nullable per pragma
reason	TEXT	NOT NULL
customer_draft	TEXT	NOT NULL
created_at	TEXT	NOT NULL
resolved_at	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
case_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_alerts_2 | UNIQUE | merchant_id, promise_id, promise_revision, day

sqlite_autoindex_platform_alerts_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_alerts (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 case_id TEXT NOT NULL REFERENCES resolution_cases(id), promise_id TEXT NOT NULL,
 promise_revision INTEGER NOT NULL, day TEXT NOT NULL, state TEXT NOT NULL,
 assignee TEXT, reason TEXT NOT NULL, customer_draft TEXT NOT NULL,
 created_at TEXT NOT NULL, resolved_at TEXT,
 UNIQUE(merchant_id,promise_id,promise_revision,day)
)
```

platform_audit

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
actor_id	TEXT	NOT NULL
action	TEXT	NOT NULL
resource_id	TEXT	nullable per pragma
recorded_at	TEXT	NOT NULL
data_json	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

platform_audit_merchant_idx | index | merchant_id, recorded_at

sqlite_autoindex_platform_audit_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_audit (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 actor_id TEXT NOT NULL, action TEXT NOT NULL, resource_id TEXT,
 recorded_at TEXT NOT NULL, data_json TEXT NOT NULL
)
```

platform_auth_limits

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
attempts	INTEGER	NOT NULL
expires_at	TEXT	NOT NULL

Indexes and uniqueness

sqlite_autoindex_platform_auth_limits_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_auth_limits (
 id TEXT PRIMARY KEY, attempts INTEGER NOT NULL, expires_at TEXT NOT NULL
)
```

platform_connections

Column	Storage type	Null / key / default
merchant_id	TEXT	NOT NULL; PK position 1
provider	TEXT	NOT NULL; PK position 2
config_json	TEXT	NOT NULL
credentials_ciphertext	TEXT	NOT NULL
enabled	INTEGER	NOT NULL
revision	INTEGER	NOT NULL
checked_at	TEXT	nullable per pragma
check_status	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_connections_1 | UNIQUE | merchant_id, provider

Stored table definition

```
CREATE TABLE platform_connections (
 merchant_id TEXT NOT NULL REFERENCES merchants(id), provider TEXT NOT NULL,
 config_json TEXT NOT NULL, credentials_ciphertext TEXT NOT NULL,
 enabled INTEGER NOT NULL CHECK(enabled IN (0,1)), revision INTEGER NOT NULL,
 checked_at TEXT, check_status TEXT NOT NULL, updated_at TEXT NOT NULL,
 PRIMARY KEY(merchant_id,provider)
)
```

platform_imports

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
provider	TEXT	NOT NULL
ticket_id	TEXT	NOT NULL
case_id	TEXT	nullable per pragma
order_reference	TEXT	nullable per pragma
title	TEXT	NOT NULL
state	TEXT	NOT NULL
suggestions_json	TEXT	nullable per pragma
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
case_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_imports_2 | UNIQUE | merchant_id, provider, ticket_id

sqlite_autoindex_platform_imports_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_imports (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 provider TEXT NOT NULL, ticket_id TEXT NOT NULL, case_id TEXT REFERENCES resolution_cases(id),
 order_reference TEXT, title TEXT NOT NULL, state TEXT NOT NULL,
 suggestions_json TEXT, created_at TEXT NOT NULL, updated_at TEXT NOT NULL,
 UNIQUE(merchant_id,provider,ticket_id)
)
```

platform_invites

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
email	TEXT	NOT NULL
role	TEXT	NOT NULL
token_hash	TEXT	NOT NULL
expires_at	TEXT	NOT NULL
used_at	TEXT	nullable per pragma
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_invites_2 | UNIQUE | token_hash

sqlite_autoindex_platform_invites_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_invites (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 email TEXT NOT NULL, role TEXT NOT NULL CHECK(role IN ('manager','finance','agent','viewer')),
 token_hash TEXT NOT NULL UNIQUE, expires_at TEXT NOT NULL, used_at TEXT,
 created_at TEXT NOT NULL
)
```

platform_offers

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
revision	INTEGER	NOT NULL
choices_json	TEXT	NOT NULL
expires_at	TEXT	NOT NULL
status	TEXT	NOT NULL
selected_choice	TEXT	nullable per pragma
decision_at	TEXT	nullable per pragma
decision_actor	TEXT	nullable per pragma
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
case_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_offers_2 | UNIQUE | case_id, revision

sqlite_autoindex_platform_offers_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_offers (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 case_id TEXT NOT NULL REFERENCES resolution_cases(id), revision INTEGER NOT NULL,
 choices_json TEXT NOT NULL, expires_at TEXT NOT NULL, status TEXT NOT NULL,
 selected_choice TEXT, decision_at TEXT, decision_actor TEXT,
 created_at TEXT NOT NULL, UNIQUE(case_id,revision)
)
```

platform_packages

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
dispute_id	TEXT	NOT NULL
reason	TEXT	NOT NULL
snapshot_json	TEXT	NOT NULL
snapshot_hash	TEXT	NOT NULL
created_at	TEXT	NOT NULL
reviewed_at	TEXT	nullable per pragma
reviewer_id	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
case_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_packages_2 | UNIQUE | merchant_id, case_id, snapshot_hash

sqlite_autoindex_platform_packages_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_packages (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 case_id TEXT NOT NULL REFERENCES resolution_cases(id), dispute_id TEXT NOT NULL,
 reason TEXT NOT NULL, snapshot_json TEXT NOT NULL, snapshot_hash TEXT NOT NULL,
 created_at TEXT NOT NULL, reviewed_at TEXT, reviewer_id TEXT,
 UNIQUE(merchant_id,case_id,snapshot_hash)
)
```

platform_returns

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
order_id	TEXT	NOT NULL
request_id	TEXT	NOT NULL
revision	INTEGER	NOT NULL
status	TEXT	NOT NULL
items_json	TEXT	NOT NULL
reason	TEXT	NOT NULL
tracking_json	TEXT	nullable per pragma
replacement_json	TEXT	nullable per pragma
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
order_id	commerce_orders.id	NO ACTION / NO ACTION
case_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_returns_2 | UNIQUE | merchant_id, request_id

sqlite_autoindex_platform_returns_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_returns (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 case_id TEXT NOT NULL REFERENCES resolution_cases(id), order_id TEXT NOT NULL REFERENCES commerce_orders(id),
 request_id TEXT NOT NULL, revision INTEGER NOT NULL, status TEXT NOT NULL,
 items_json TEXT NOT NULL, reason TEXT NOT NULL, tracking_json TEXT,
 replacement_json TEXT, created_at TEXT NOT NULL, updated_at TEXT NOT NULL,
 UNIQUE(merchant_id,request_id)
)
```

platform_sessions

Column	Storage type	Null / key / default
token_hash	TEXT	nullable per pragma; PK position 1
staff_id	TEXT	NOT NULL
created_at	TEXT	NOT NULL
expires_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
staff_id	platform_staff.id	NO ACTION / NO ACTION

Indexes and uniqueness

platform_sessions_staff_idx | index | staff_id

sqlite_autoindex_platform_sessions_1 | UNIQUE | token_hash

Stored table definition

```
CREATE TABLE platform_sessions (
 token_hash TEXT PRIMARY KEY, staff_id TEXT NOT NULL REFERENCES platform_staff(id),
 created_at TEXT NOT NULL, expires_at TEXT NOT NULL
)
```

platform_staff

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
email	TEXT	NOT NULL
name	TEXT	NOT NULL
password_hash	TEXT	NOT NULL
role	TEXT	NOT NULL
active	INTEGER	NOT NULL; default 1
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_staff_2 | UNIQUE | email

sqlite_autoindex_platform_staff_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_staff (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 email TEXT NOT NULL UNIQUE, name TEXT NOT NULL, password_hash TEXT NOT NULL,
 role TEXT NOT NULL CHECK(role IN ('owner','manager','finance','agent','viewer')),
 active INTEGER NOT NULL DEFAULT 1 CHECK(active IN (0,1)), created_at TEXT NOT NULL
)
```

platform_ticket_messages

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
import_id	TEXT	NOT NULL
source_id	TEXT	NOT NULL
sender	TEXT	NOT NULL
body	TEXT	NOT NULL
public	INTEGER	NOT NULL
sent_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
import_id	platform_imports.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_ticket_messages_2 | UNIQUE | import_id, source_id

sqlite_autoindex_platform_ticket_messages_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_ticket_messages (
 id TEXT PRIMARY KEY, import_id TEXT NOT NULL REFERENCES platform_imports(id),
 source_id TEXT NOT NULL, sender TEXT NOT NULL CHECK(sender IN ('buyer','merchant')),
 body TEXT NOT NULL, public INTEGER NOT NULL CHECK(public IN (0,1)), sent_at TEXT NOT NULL,
 UNIQUE(import_id,source_id)
)
```

platform_webhook_jobs

Column	Storage type	Null / key / default
id	TEXT	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
provider	TEXT	NOT NULL
event_id	TEXT	NOT NULL
body_hash	TEXT	NOT NULL
ticket_id	TEXT	NOT NULL
state	TEXT	NOT NULL
attempts	INTEGER	NOT NULL; default 0
next_attempt_at	TEXT	NOT NULL
lease_id	TEXT	nullable per pragma
error	TEXT	nullable per pragma
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_platform_webhook_jobs_2 | UNIQUE | merchant_id, provider, event_id

sqlite_autoindex_platform_webhook_jobs_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE platform_webhook_jobs (
 id TEXT PRIMARY KEY, merchant_id TEXT NOT NULL REFERENCES merchants(id),
 provider TEXT NOT NULL, event_id TEXT NOT NULL, body_hash TEXT NOT NULL,
 ticket_id TEXT NOT NULL, state TEXT NOT NULL, attempts INTEGER NOT NULL DEFAULT 0,
 next_attempt_at TEXT NOT NULL, lease_id TEXT, error TEXT, created_at TEXT NOT NULL,
 UNIQUE(merchant_id,provider,event_id)
)
```

provider_refresh_schedule

Column	Storage type	Null / key / default
promise_id	TEXT	NOT NULL; PK position 1
revision	INTEGER	NOT NULL
next_attempt_at	TEXT	NOT NULL
failures	INTEGER	NOT NULL; default 0
lease_id	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
promise_id	resolution_promises.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_provider_refresh_schedule_1 | UNIQUE | promise_id

Stored table definition

```
CREATE TABLE provider_refresh_schedule (
    promise_id TEXT PRIMARY KEY NOT NULL REFERENCES resolution_promises(id),
    revision INTEGER NOT NULL,
    next_attempt_at TEXT NOT NULL,
    failures INTEGER NOT NULL DEFAULT 0,
    lease_id TEXT
)
```

refund_approval_events

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
approval_id	TEXT	NOT NULL
event_type	TEXT	NOT NULL
actor_merchant_id	TEXT	nullable per pragma
note	TEXT	NOT NULL
recorded_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
actor_merchant_id	merchants.id	NO ACTION / NO ACTION
approval_id	refund_approvals.id	NO ACTION / NO ACTION

Indexes and uniqueness

approval_events_approval_idx | index | approval_id, sequence

Stored table definition

```
CREATE TABLE refund_approval_events (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    approval_id TEXT NOT NULL REFERENCES refund_approvals(id),
    event_type TEXT NOT NULL CHECK (event_type IN ('approved', 'revoked', 'expired', 'invalidated')),
    actor_merchant_id TEXT REFERENCES merchants(id),
    note TEXT NOT NULL,
    recorded_at TEXT NOT NULL
)
```

refund_approvals

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
request_id	TEXT	NOT NULL
amount_cents	INTEGER	NOT NULL
currency	TEXT	NOT NULL
state_fingerprint	TEXT	NOT NULL
request_json	TEXT	NOT NULL
snapshot_json	TEXT	NOT NULL
decision_note	TEXT	NOT NULL
status	TEXT	NOT NULL
approved_at	TEXT	NOT NULL
expires_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL
status_reason	TEXT	NOT NULL
revocation_reason	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
capture_id	cases.capture_id	CASCADE / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

approval_execution_owner_idx | UNIQUE | id, merchant_id, case_id, capture_id

approvals_case_idx | index | case_id, approved_at

one_active_approval_per_capture | UNIQUE | merchant_id, capture_id

sqlite_autoindex_refund_approvals_2 | UNIQUE | merchant_id, request_id

sqlite_autoindex_refund_approvals_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE refund_approvals (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    case_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    request_id TEXT NOT NULL,
    amount_cents INTEGER NOT NULL CHECK (amount_cents > 0),
    currency TEXT NOT NULL CHECK (currency = 'USD'),
    state_fingerprint TEXT NOT NULL,
    request_json TEXT NOT NULL,
    snapshot_json TEXT NOT NULL,
    decision_note TEXT NOT NULL,
    status TEXT NOT NULL CHECK (status IN ('approved', 'revoked', 'expired', 'invalidated')),
    approved_at TEXT NOT NULL,
    expires_at TEXT NOT NULL,
    updated_at TEXT NOT NULL,
    status_reason TEXT NOT NULL,
    revocation_reason TEXT,
    UNIQUE (merchant_id, request_id),
    FOREIGN KEY (case_id, merchant_id, capture_id)
        REFERENCES cases(id, merchant_id, capture_id) ON UPDATE CASCADE
)
```

refund_execution_events

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
execution_id	TEXT	NOT NULL
status	TEXT	NOT NULL
note	TEXT	NOT NULL
recorded_at	TEXT	NOT NULL
snapshot_json	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
execution_id	refund_executions.id	NO ACTION / NO ACTION

Stored table definition

```
CREATE TABLE refund_execution_events (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    execution_id TEXT NOT NULL REFERENCES refund_executions(id),
    status TEXT NOT NULL CHECK (status IN ('reserved','submitting','unknown','pending','completed','failed')),
    note TEXT NOT NULL,
    recorded_at TEXT NOT NULL,
    snapshot_json TEXT
)
```

refund_executions

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
approval_id	TEXT	NOT NULL
request_id	TEXT	NOT NULL
provider_request_id	TEXT	NOT NULL
amount_cents	INTEGER	NOT NULL
currency	TEXT	NOT NULL
status	TEXT	NOT NULL
attempt_count	INTEGER	NOT NULL; default 0
approval_json	TEXT	NOT NULL
capture_check_json	TEXT	NOT NULL
provider_request_json	TEXT	NOT NULL
provider_refund_id	TEXT	nullable per pragma
provider_refund_json	TEXT	nullable per pragma
error_code	TEXT	nullable per pragma
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL
lease_expires_at	TEXT	nullable per pragma
provider_identity_hash	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
approval_id	refund_approvals.id	CASCADE / NO ACTION
merchant_id	refund_approvals.merchant_id	CASCADE / NO ACTION
case_id	refund_approvals.case_id	CASCADE / NO ACTION
capture_id	refund_approvals.capture_id	CASCADE / NO ACTION
case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
capture_id	cases.capture_id	CASCADE / NO ACTION

Indexes and uniqueness

refund_executions_active_capture_idx | UNIQUE | merchant_id, capture_id

refund_executions_case_idx | index | case_id, created_at

sqlite_autoindex_refund_executions_4 | UNIQUE | merchant_id, request_id

sqlite_autoindex_refund_executions_3 | UNIQUE | provider_request_id

sqlite_autoindex_refund_executions_2 | UNIQUE | approval_id

sqlite_autoindex_refund_executions_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE refund_executions (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    case_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    approval_id TEXT NOT NULL UNIQUE,
    request_id TEXT NOT NULL,
    provider_request_id TEXT NOT NULL UNIQUE,
    amount_cents INTEGER NOT NULL CHECK (amount_cents > 0),
    currency TEXT NOT NULL CHECK (currency = 'USD'),
    status TEXT NOT NULL CHECK (status IN ('reserved','submitting','unknown','pending','completed','failed')),
    attempt_count INTEGER NOT NULL DEFAULT 0 CHECK (attempt_count >= 0),
    approval_json TEXT NOT NULL,
    capture_check_json TEXT NOT NULL,
    provider_request_json TEXT NOT NULL,
    provider_refund_id TEXT,
    provider_refund_json TEXT,
    error_code TEXT,
    created_at TEXT NOT NULL,
    updated_at TEXT NOT NULL,
    lease_expires_at TEXT,
    provider_identity_hash TEXT,
    UNIQUE (merchant_id, request_id),
    FOREIGN KEY (case_id, merchant_id, capture_id)
        REFERENCES cases(id, merchant_id, capture_id) ON UPDATE CASCADE,
    FOREIGN KEY (approval_id, merchant_id, case_id, capture_id)
        REFERENCES refund_approvals(id, merchant_id, case_id, capture_id) ON UPDATE CASCADE
)
```

refunds

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
merchant_id	TEXT	NOT NULL
refund_id	TEXT	NOT NULL
capture_id	TEXT	NOT NULL
amount_cents	INTEGER	NOT NULL
currency	TEXT	NOT NULL
status	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
merchant_id	captures.merchant_id	CASCADE / NO ACTION
capture_id	captures.capture_id	CASCADE / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

refunds_merchant_capture_idx | index | merchant_id, capture_id

sqlite_autoindex_refunds_1 | UNIQUE | merchant_id, refund_id

Stored table definition

```
CREATE TABLE refunds (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    refund_id TEXT NOT NULL,
    capture_id TEXT NOT NULL,
    amount_cents INTEGER NOT NULL CHECK (amount_cents > 0),
    currency TEXT NOT NULL CHECK (currency = 'USD'),
    status TEXT NOT NULL CHECK (status IN ('pending', 'completed', 'failed')),
    UNIQUE (merchant_id, refund_id),
    FOREIGN KEY (merchant_id, capture_id)
        REFERENCES captures(merchant_id, capture_id) ON UPDATE CASCADE
)
```

resolution_cases

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
refund_case_id	TEXT	nullable per pragma
title	TEXT	NOT NULL
order_reference	TEXT	nullable per pragma
created_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
refund_case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

resolution_cases_owner_idx | index | merchant_id, created_at, id

sqlite_autoindex_resolution_cases_3 | UNIQUE | merchant_id, refund_case_id

sqlite_autoindex_resolution_cases_2 | UNIQUE | id, merchant_id

sqlite_autoindex_resolution_cases_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE resolution_cases (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    refund_case_id TEXT,
    title TEXT NOT NULL,
    order_reference TEXT,
    created_at TEXT NOT NULL,
    UNIQUE (id, merchant_id),
    UNIQUE (merchant_id, refund_case_id),
    FOREIGN KEY (refund_case_id, merchant_id) REFERENCES cases(id, merchant_id) ON UPDATE CASCADE
)
```

resolution_events

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
id	TEXT	NOT NULL
resolution_id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
promise_id	TEXT	nullable per pragma
revision	INTEGER	nullable per pragma
event_type	TEXT	NOT NULL
source	TEXT	NOT NULL
recorded_at	TEXT	NOT NULL
data_json	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
promise_id	resolution_promises.id	CASCADE / NO ACTION
resolution_id	resolution_promises.resolution_id	CASCADE / NO ACTION
merchant_id	resolution_promises.merchant_id	CASCADE / NO ACTION
resolution_id	resolution_cases.id	CASCADE / NO ACTION
merchant_id	resolution_cases.merchant_id	CASCADE / NO ACTION

Indexes and uniqueness

resolution_events_proof_idx | index | promise_id, revision, sequence

resolution_events_case_idx | index | resolution_id, sequence

sqlite_autoindex_resolution_events_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE resolution_events (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    id TEXT NOT NULL UNIQUE,
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    promise_id TEXT,
    revision INTEGER,
    event_type TEXT NOT NULL CHECK (event_type IN ('case_created','promise_recorded','promise_revised',
        'merchant_evidence','provider_observation','verification_failed')),
    source TEXT NOT NULL CHECK (source IN ('merchant_recorded','provider_confirmed','system')),
    recorded_at TEXT NOT NULL,
    data_json TEXT NOT NULL,
    CHECK ((promise_id IS NULL) = (revision IS NULL)),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) ON UPDATE CASCADE,
    FOREIGN KEY (promise_id, resolution_id, merchant_id)
        REFERENCES resolution_promises(id, resolution_id, merchant_id) ON UPDATE CASCADE
)
```

resolution_evidence_events

Column	Storage type	Null / key / default
resolution_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
origin	TEXT	NOT NULL; PK position 2
source	TEXT	NOT NULL; PK position 3
event_key	TEXT	NOT NULL; PK position 4
payload_hash	TEXT	NOT NULL
event_id	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION
event_id	resolution_events.id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_resolution_evidence_events_2 | UNIQUE | resolution_id, origin, source, event_key

sqlite_autoindex_resolution_evidence_events_1 | UNIQUE | event_id

Stored table definition

```
CREATE TABLE resolution_evidence_events (
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    origin TEXT NOT NULL CHECK (origin IN ('merchant','paypal')),
    source TEXT NOT NULL,
    event_key TEXT NOT NULL,
    payload_hash TEXT NOT NULL,
    event_id TEXT NOT NULL UNIQUE REFERENCES resolution_events(id) DEFERRABLE INITIALLY IMMEDIATE,
    PRIMARY KEY (resolution_id, origin, source, event_key),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) DEFERRABLE INITIALLY IMMEDIATE
)
```

resolution_evidence_links

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
resolution_id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
kind	TEXT	NOT NULL
reference	TEXT	NOT NULL
source	TEXT	NOT NULL
linked_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

resolution_evidence_primary_payment | UNIQUE | resolution_id, kind

sqlite_autoindex_resolution_evidence_links_2 | UNIQUE | resolution_id, kind, reference

sqlite_autoindex_resolution_evidence_links_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE resolution_evidence_links (
    id TEXT PRIMARY KEY NOT NULL,
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    kind TEXT NOT NULL CHECK (kind IN ('order','paypal_order','paypal_capture','paypal_refund','support_ticket')),
    reference TEXT NOT NULL,
    source TEXT NOT NULL,
    linked_at TEXT NOT NULL,
    UNIQUE (resolution_id, kind, reference),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) DEFERRABLE INITIALLY IMMEDIATE
)
```

resolution_evidence_refs

Column	Storage type	Null / key / default
merchant_id	TEXT	NOT NULL; PK position 1
external_reference	TEXT	NOT NULL; PK position 2
resolution_id	TEXT	NOT NULL
request_hash	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_resolution_evidence_refs_2 | UNIQUE | merchant_id, external_reference

sqlite_autoindex_resolution_evidence_refs_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE resolution_evidence_refs (
    merchant_id TEXT NOT NULL,
    external_reference TEXT NOT NULL,
    resolution_id TEXT NOT NULL UNIQUE,
    request_hash TEXT NOT NULL,
    PRIMARY KEY (merchant_id, external_reference),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) DEFERRABLE INITIALLY IMMEDIATE
)
```

resolution_evidence_state

Column	Storage type	Null / key / default
resolution_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
started_at	TEXT	NOT NULL
data_json	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_resolution_evidence_state_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE resolution_evidence_state (
    resolution_id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    started_at TEXT NOT NULL,
    data_json TEXT NOT NULL,
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) DEFERRABLE INITIALLY IMMEDIATE
)
```

resolution_helpdesk

Column	Storage type	Null / key / default
resolution_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
host	TEXT	NOT NULL
ticket_id	INTEGER	NOT NULL
status	TEXT	NOT NULL
checked_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	NO ACTION / NO ACTION
merchant_id	resolution_cases.merchant_id	NO ACTION / NO ACTION

Indexes and uniqueness

sqlite_autoindex_resolution_helpdesk_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE resolution_helpdesk (
    resolution_id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    host TEXT NOT NULL,
    ticket_id INTEGER NOT NULL,
    status TEXT NOT NULL,
    checked_at TEXT NOT NULL,
    FOREIGN KEY (resolution_id,merchant_id) REFERENCES resolution_cases(id,merchant_id)
)
```

resolution_operations

Column	Storage type	Null / key / default
resolution_id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
assignee	TEXT	nullable per pragma
due_at	TEXT	nullable per pragma
waiting_on_customer	INTEGER	NOT NULL; default 0
read_through	INTEGER	NOT NULL; default 0
revision	INTEGER	NOT NULL; default 0

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	CASCADE / NO ACTION
merchant_id	resolution_cases.merchant_id	CASCADE / NO ACTION

Indexes and uniqueness

sqlite_autoindex_resolution_operations_1 | UNIQUE | resolution_id

Stored table definition

```
CREATE TABLE resolution_operations (
    resolution_id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL,
    assignee TEXT,
    due_at TEXT,
    waiting_on_customer INTEGER NOT NULL DEFAULT 0 CHECK(waiting_on_customer IN (0,1)),
    read_through INTEGER NOT NULL DEFAULT 0 CHECK(read_through >= 0),
    revision INTEGER NOT NULL DEFAULT 0 CHECK(revision >= 0),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) ON UPDATE CASCADE
)
```

resolution_promises

Column	Storage type	Null / key / default
sequence	INTEGER	nullable per pragma; PK position 1
id	TEXT	NOT NULL
resolution_id	TEXT	NOT NULL
merchant_id	TEXT	NOT NULL
kind	TEXT	NOT NULL
revision	INTEGER	NOT NULL
promise_json	TEXT	NOT NULL
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Foreign keys

Local column	Target	Update / delete
resolution_id	resolution_cases.id	CASCADE / NO ACTION
merchant_id	resolution_cases.merchant_id	CASCADE / NO ACTION

Indexes and uniqueness

resolution_promises_case_idx | index | resolution_id, sequence

resolution_refund_promise_idx | UNIQUE | resolution_id

sqlite_autoindex_resolution_promises_2 | UNIQUE | id, resolution_id, merchant_id

sqlite_autoindex_resolution_promises_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE resolution_promises (
    sequence INTEGER PRIMARY KEY AUTOINCREMENT,
    id TEXT NOT NULL UNIQUE,
    resolution_id TEXT NOT NULL,
    merchant_id TEXT NOT NULL,
    kind TEXT NOT NULL CHECK (kind IN ('refund','return','replacement','order_cancellation',
        'subscription','store_credit_or_exchange','billing_correction','dispute_handoff')),
    revision INTEGER NOT NULL CHECK (revision > 0),
    promise_json TEXT NOT NULL,
    created_at TEXT NOT NULL,
    updated_at TEXT NOT NULL,
    UNIQUE (id, resolution_id, merchant_id),
    FOREIGN KEY (resolution_id, merchant_id) REFERENCES resolution_cases(id, merchant_id) ON UPDATE CASCADE
)
```

review_exceptions

Column	Storage type	Null / key / default
id	TEXT	NOT NULL; PK position 1
merchant_id	TEXT	NOT NULL
case_id	TEXT	nullable per pragma
capture_id	TEXT	nullable per pragma
source_key	TEXT	NOT NULL
reason	TEXT	NOT NULL
status	TEXT	NOT NULL
blocking	INTEGER	NOT NULL
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL
resolution_note	TEXT	nullable per pragma

Foreign keys

Local column	Target	Update / delete
case_id	cases.id	CASCADE / NO ACTION
merchant_id	cases.merchant_id	CASCADE / NO ACTION
capture_id	cases.capture_id	CASCADE / NO ACTION
merchant_id	merchants.id	NO ACTION / NO ACTION

Indexes and uniqueness

review_exceptions_capture_idx | index | merchant_id, capture_id, status, blocking

review_exceptions_queue_idx | index | merchant_id, status, updated_at

sqlite_autoindex_review_exceptions_2 | UNIQUE | merchant_id, source_key

sqlite_autoindex_review_exceptions_1 | UNIQUE | id

Stored table definition

```
CREATE TABLE review_exceptions (
    id TEXT PRIMARY KEY NOT NULL,
    merchant_id TEXT NOT NULL REFERENCES merchants(id),
    case_id TEXT,
    capture_id TEXT,
    source_key TEXT NOT NULL,
    reason TEXT NOT NULL,
    status TEXT NOT NULL CHECK (status IN ('open','resolved')),
    blocking INTEGER NOT NULL CHECK (blocking IN (0,1)),
    created_at TEXT NOT NULL,
    updated_at TEXT NOT NULL,
    resolution_note TEXT,
    UNIQUE (merchant_id, source_key),
    FOREIGN KEY (case_id, merchant_id, capture_id)
        REFERENCES cases(id, merchant_id, capture_id) ON UPDATE CASCADE
)
```

Reading map

This volume brings together the maintained source documents listed below. Source chapters retain their recorded evidence dates and assumptions. Later corrections in the engineering log take precedence over older snapshots.

Chapter	Source in the project
PostgreSQL migration procedure	docs/POSTGRES_MIGRATION.md

PostgreSQL migration procedure

Source: docs/POSTGRES_MIGRATION.md

Move Resolve from SQLite to PostgreSQL

Choose **Neon** for the current Resolve demo. We need a PostgreSQL database; FastAPI already owns merchant access, customer sign-in, and evidence storage. The repository uses ordinary PostgreSQL with no vendor SDK, so Supabase is also an option if you want its Auth and Storage platform (https://supabase.com/docs/guides/database/overview) for later milestones. The migration itself does not switch either service.

Import into a fresh database

1. Create a database or disposable branch in Neon (https://console.neon.tech/).

Copy the **direct** connection URL from the Connect dialog for the import. Use the provider's TLS options. For Supabase, use its direct URL or the session pooler (https://supabase.com/docs/guides/database/connecting-to-postgres) if your machine cannot reach the direct host.

2. Stop the SQLite API and both workers before the final copy. Keep a backup

of the source file. Put the connection string in `backend/.env` (ignored by Git). Use a dedicated Resolve database or branch; the app owns a private `resolve` schema. Do not put a database URL in frontend settings or chat.

```
RESOLVE_DATABASE_URL=postgresql://ROLE:PASSWORD@HOST/DATABASE?sslmode=require
```

3. From the repository root, run:

```
uv --directory backend sync
uv --directory backend run python -m app.migrate_sqlite
```

The default source is the configured SQLite file. Use `--source PATH` to import a specific backup. The importer opens SQLite read-only, upgrades a private copy if needed, checks its integrity, and copies to an empty Resolve schema. It verifies foreign keys and compares the copied rows, including attachment bytes, before committing. It keeps record IDs, stored key hashes, timestamps, cents, and identity sequence high-water marks. A populated destination is rejected; there is no merge mode.

4. Start FastAPI with the same `RESOLVE_DATABASE_URL`. Verify an existing

merchant key, case, and settlement before starting the workers. If you later switch the runtime to Neon's pooled endpoint (https://neon.com/docs/connect/choose-connection), select the same branch and database. The API and both workers must use the same database. Inspect any imported pending work before enabling delivery or refund execution.

SQLite remains the default when `RESOLVE_DATABASE_URL` is unset. The import does not change the source file or the app's environment. Once writes start on PostgreSQL, do not switch back to the old SQLite copy as a rollback without accounting for writes made since the copy.

The API contract is unchanged. PostgreSQL writes take a transaction-scoped lock in the Resolve schema before reading or changing state, preserving the SQLite serialization used by refund approvals and reservations. Reads use a repeatable snapshot. This first implementation prioritizes correctness and works with one writer at a time; it does not claim high-volume write scaling. SQLite and PostgreSQL have separate migration histories; add corresponding schema changes to both while both backends are supported. The database stores evidence as bytes, so monitor storage as uploads grow. If you use a free Neon tier, check its usage limits (https://neon.com/docs/introduction/plans) when running an always-on polling worker.

PostgreSQL migration history errors

Startup compares the applied migration history to the SQL shipped with Resolve. The original migration 003 is preserved, and migration 004 adds the PayPal callback claim column. The original migration 005 is also preserved; 006 adds email notification flags to existing invites and contacts. Resolve recognizes the exact interim checksum for each migration if the new columns were already applied. Restart the backend with the updated code and the same database configuration to upgrade.

If an unknown checksum or a newer schema still prevents startup, check that the running release has the original migrations for that database. Keep the history table and existing data intact; editing checksums or resetting the schema does not repair a migration mismatch.

Treat numbered migrations as immutable even before committing: a backend running with `--reload` may already have applied them. The default backend test suite checks the known PostgreSQL migration checksums even without a PostgreSQL connection.

Test PostgreSQL locally

Point the integration tests to a disposable PostgreSQL database. They create and remove uniquely named test schemas. Use a separate database from the one you plan to deploy.

```
$env:RESOLVE_TEST_DATABASE_URL = 'postgresql://USER:PASSWORD@127.0.0.1:5432/TEST_DB'
$env:RESOLVE_TEST_STORAGE = 'postgres'
uv --directory backend run python -m pytest -q tests/test_postgres.py tests/test_customer_portal.py
tests/test_customer_workflow.py tests/test_hosted_followup.py tests/test_resolution_workspace.py
```